

Self-Check Questions: Miscellaneous Topics

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1 Riemann and Darboux Integration

1.1 Foundational Definitions

1. What is a partition of an interval $[a, b]$?
2. Define the norm (mesh size) of a partition.
3. What is a tagged partition?
4. Define a Riemann sum.
5. State the definition of the Riemann integral using Riemann sums.
6. Define the upper Darboux sum.
7. Define the lower Darboux sum.
8. What are the quantities

$$M_i = \sup_{x \in [x_{i-1}, x_i]} f(x)$$

and

$$m_i = \inf_{x \in [x_{i-1}, x_i]} f(x)?$$

9. Define the upper Darboux integral.
10. Define the lower Darboux integral.
11. State the definition of Darboux integrability.
12. Why must a function be bounded to be Riemann integrable?
13. Why must a function be bounded to be Darboux integrable?

1.2 Conceptual Understanding

14. What is the geometric intuition behind a Riemann sum?
15. What is the geometric intuition behind upper and lower Darboux sums?
16. Why does refining a partition generally improve approximation quality?
17. What happens to lower Darboux sums when a partition is refined?
18. What happens to upper Darboux sums when a partition is refined?
19. Why is

$$L(f, P) \leq U(f, P)$$

always true?

20. Explain why

$$\sup_P L(f, P) \leq \inf_P U(f, P).$$

21. What does it mean geometrically when the upper and lower integrals are equal?
22. Why are Darboux sums often considered technically simpler than Riemann sums?
23. Why are Riemann sums closer to the historical intuition of integration?

1.3 Comparisons Between Riemann and Darboux Integration

24. What is the main difference between a Riemann sum and a Darboux sum?
25. In a Riemann sum, how are sample points chosen?
26. In a Darboux sum, why are sample points unnecessary?
27. Which theory uses arbitrary sample points explicitly?
28. Which theory uses suprema and infima on subintervals?
29. What role does the partition norm play in Riemann integration?
30. What role does partition refinement play in Darboux integration?
31. State the theorem relating Riemann integrability and Darboux integrability.
32. Why are the two theories equivalent for bounded functions on closed intervals?
33. Can two different tagged partitions with the same underlying partition produce different Riemann sums?
34. Can two different Darboux sums exist for the same partition?
35. How do upper and lower Darboux sums bound all possible Riemann sums?
36. Explain how Darboux integration avoids explicit limits of sums.
37. Explain how Riemann integration encodes convergence directly through limits.

1.4 Theorems and Properties

38. State the linearity property of the integral.
39. State the additivity over intervals property.
40. State the monotonicity property of the integral.
41. If f is continuous on $[a, b]$, why is it integrable?
42. Why is every monotone function on $[a, b]$ integrable?
43. State a criterion for integrability involving discontinuities.
44. What is the relationship between measure-zero discontinuity sets and integrability?
45. Why is the Dirichlet function not Riemann integrable?
46. Why is the characteristic function of the rationals discontinuous everywhere?
47. Give an example of a function that is discontinuous but still Riemann integrable.
48. Why is a function with finitely many discontinuities integrable?
49. What happens to the integral if two functions differ only on finitely many points?
50. What happens if two functions differ only on a measure-zero set?

1.5 Technical Manipulation

51. Compute the lower and upper Darboux sums for a constant function.
52. Compute the lower and upper Darboux sums for $f(x) = x$ on $[0, 1]$ using a uniform partition.

53. Derive the Riemann sum for $f(x) = x^2$ on $[0, 1]$.

54. Show directly that

$$\int_0^1 x \, dx = \frac{1}{2}$$

using Riemann sums.

55. Show directly that

$$\int_0^1 x \, dx = \frac{1}{2}$$

using Darboux sums.

56. For a uniform partition, how do you compute

$$\Delta x?$$

57. Why do continuous functions attain maxima and minima on closed intervals?

58. Why is compactness important in Darboux integration?

59. What is the difference between taking a supremum over all lower sums and taking a limit?

60. Why does the existence of arbitrarily close upper and lower sums imply integrability?

1.6 Interpreting the Integral

61. What does the expression

$$\sum_i f(x_i) \Delta x_i$$

represent before assigning any geometric or physical interpretation to it?

62. Why is the integral fundamentally an accumulation operator rather than specifically an area operator?

63. In the area interpretation of the integral, what do the quantities

$$f(x_i), \quad \Delta x_i, \quad f(x_i) \Delta x_i$$

represent geometrically?

64. Why does

$$f(x_i) \Delta x_i$$

approximate the area of a thin rectangle?

65. What happens to the Riemann sum as

$$\max \Delta x_i \rightarrow 0?$$

66. Why does the definite integral emerge as a limit of Riemann sums?

67. If

$$g'(x) = f(x),$$

why does

$$f(x) \Delta x$$

approximate the small change

$$\Delta g?$$

68. What is the conceptual meaning of

$$dg = g'(x) dx?$$

69. Why can the same expression

$$f(x)\Delta x$$

be interpreted both as rectangle area and as accumulated change?

70. What determines the interpretation of the integrand $f(x)$?

71. Why is the equality

$$\int_a^b g'(x) dx = g(b) - g(a)$$

an accumulation statement?

72. In the Fundamental Theorem of Calculus, what exactly is being accumulated?

73. Why does integrating a derivative recover net change?

74. How does the telescoping intuition appear in

$$\sum \Delta g_i?$$

75. Define

$$A(x) = \int_a^x f(t) dt.$$

What does $A(x)$ represent geometrically?

76. Why does increasing x by a small amount h add approximately

$$f(x)h$$

to the accumulated area?

77. Use the difference quotient to explain why

$$A'(x) = f(x).$$

78. Why does the identity

$$A'(x) = f(x)$$

connect the area interpretation to the antiderivative interpretation?

79. Why is the Fundamental Theorem of Calculus fundamentally a statement that differentiation and integration are inverse operations?

80. How does the notion of “local rate” connect to “global accumulation”?

81. What is the difference between:

- a function value,
- a derivative,
- a differential,
- an accumulated quantity?

82. Why is

$$\int_a^b f(x) dx$$

not intrinsically an area?

83. Give examples where the integral represents:

- distance,
- mass,
- charge,
- probability,
- energy.

84. In each example above, what plays the role of:

- density/rate,
- small measure,
- accumulated total?

85. Explain the pattern:

$$\text{density} \times \text{small measure} \rightarrow \text{accumulated quantity}.$$

86. Why is the rectangle picture only one geometric realization of integration?

87. What is the conceptual difference between:

$$f(x) \quad \text{and} \quad f(x) dx?$$

88. Why is

$$dx$$

best understood as “a tiny increment of the independent variable”?

89. Why is the statement

$$dg = g'(x) dx$$

more fundamental than the rectangle interpretation?

90. What does the Fundamental Theorem of Calculus reveal about the relationship between local structure and global structure?

1.7 Integrals and Euler’s Method

91. Consider the initial value problem

$$y'(x) = f(x, y(x)), \quad y(a) = y_0.$$

Write down the forward Euler update on a partition

$$a = x_0 < x_1 < \cdots < x_n = b$$

and explain why one step has the form

$$y_{i+1} - y_i \approx f(x_i, y_i) \Delta x_i.$$

92. Why is a single Euler update naturally interpreted as an accumulation of a local rate over a small interval?

93. Explain why it is reasonable to call

$$y_{i+1} = y_i + f(x_i, y_i) \Delta x_i$$

an integration step rather than merely an algebraic update.

94. Sum the Euler increments over many steps and derive

$$y_n = y_0 + \sum_{i=0}^{n-1} f(x_i, y_i) \Delta x_i.$$

What is the conceptual meaning of this formula, and how does the initial value y_0 enter as the starting accumulated amount?

95. Why does the exact solution of

$$y'(x) = f(x, y(x)), \quad y(a) = y_0,$$

satisfy the integral equation

$$y(x) = y_0 + \int_a^x f(t, y(t)) dt?$$

How does Euler's method mimic this identity?

96. Why is the solution curve of an ODE often called an integral curve? What is being integrated, and in what sense is the curve determined by that integration?

97. For a general first-order ODE

$$y' = f(x, y),$$

what information does the slope field encode? Why does prescribing an initial value

$$y(a) = y_0$$

select one particular curve from that field?

98. In the autonomous case

$$y' = F(y),$$

how does the interpretation of an integral curve differ from the case

$$y' = f(x, y)?$$

What remains the same?

99. Show that an ordinary definite integral is the special case of the ODE

$$y'(x) = g(x), \quad y(a) = y_0,$$

where the right-hand side does not depend on y . What is the exact solution?

100. Let G be one antiderivative of g , so that

$$G'(x) = g(x).$$

Why are all solutions of

$$y'(x) = g(x)$$

of the form

$$y(x) = G(x) + C?$$

How does the initial condition

$$y(a) = y_0$$

determine the value of C ?

101. Compare the slope field of a general ODE

$$y' = f(x, y)$$

with the slope field of

$$y' = g(x).$$

Why is the latter vertically uniform, meaning that at a fixed x the slope is the same at every height y ?

102. Why does the vertical uniformity of the slope field for

$$y' = g(x)$$

imply that all integral curves are vertical translates of one another? How does this connect to the additive constant C in

$$y = G(x) + C?$$

103. Apply Euler's method to

$$y'(x) = g(x), \quad y(a) = y_0.$$

Why does the scheme become

$$y_{i+1} = y_i + g(x_i) \Delta x_i?$$

104. Iterate the previous update and show that

$$y_n = y_0 + \sum_{i=0}^{n-1} g(x_i) \Delta x_i.$$

Why is this now literally a Riemann sum plus the initial value, rather than only an Euler approximation?

105. What special feature makes the discrete accumulation for

$$y' = g(x)$$

telescope exactly into

$$y_n - y_0 = \sum_{i=0}^{n-1} (y_{i+1} - y_i)?$$

Why does the same telescoping identity not by itself turn a general Euler method for

$$y' = f(x, y)$$

into a classical Riemann sum for a known function of x alone?

106. In the case

$$y'(x) = g(x),$$

under what conditions on the partition and sample points does

$$\sum_{i=0}^{n-1} g(x_i) \Delta x_i$$

converge to

$$\int_a^b g(x) dx?$$

State the connection with the definition of Riemann integrability.

107. If

$$y'(x) = f(x, y)$$

does depend on y , what breaks in the attempt to interpret Euler's method as an ordinary Riemann sum in the variable x alone? Which additional unknown quantity is being approximated at every step?

108. Compare these two viewpoints:

$$\int_a^b g(x) dx \quad \text{and} \quad y(b) - y(a) \quad \text{for} \quad y' = f(x, y).$$

In what sense are both global accumulations of local data, and in what sense is the second problem structurally more implicit?

109. If two solutions of

$$y' = g(x)$$

have different initial values, why does their difference remain constant in x ? Why is the analogous statement usually false for a general ODE

$$y' = f(x, y)?$$

2 The Connection Between the Net Change Theorem, Differential Equations, and Euler's Method

The connection between the net change theorem, differential equations, and Euler's method is structural. They are different views of the same fundamental operation:

- Differential equations describe *local rates of change*.
- Integration accumulates those local changes into a *global change*.
- Euler's method numerically performs that accumulation step-by-step.

The net change theorem is the bridge connecting these ideas.

Suppose that

$$F'(t) = f(t).$$

Then the net change theorem states:

$$F(b) - F(a) = \int_a^b f(t) dt \quad \text{or} \quad F(b) = F(a) + \int_a^b f(t) dt.$$

The interpretation is:

- $f(t)$ is the instantaneous rate of change,
- the integral accumulates infinitesimal changes,
- the result is the total change in F .

Now consider a differential equation:

$$y'(t) = f(t, y(t)).$$

This equation states that the instantaneous change of the unknown function y is determined by the function f .

Applying the net change theorem to the unknown solution y , we integrate both sides from t_0 to t :

$$y(t) - y(t_0) = \int_{t_0}^t f(s, y(s)) \, ds.$$

If the initial value is

$$y(t_0) = y_0,$$

then:

$$y(t) = y_0 + \int_{t_0}^t f(s, y(s)) \, ds.$$

This equation has a direct interpretation:

- Start with the initial value y_0 ,
- accumulate all infinitesimal changes $f(s, y(s)) \, ds$,
- obtain the final value $y(t)$.

Thus, a differential equation is fundamentally an accumulation law.

Euler's Method as Numerical Integration

Euler's method arises by approximating the integral with a Riemann sum. Take a small step size h . Over the interval $[t_n, t_n + h]$,

$$y(t_{n+1}) - y(t_n) = \int_{t_n}^{t_n+h} f(s, y(s)) \, ds.$$

Euler's approximation assumes that the slope remains approximately constant over this short interval:

$$f(s, y(s)) \approx f(t_n, y_n).$$

Therefore,

$$\int_{t_n}^{t_n+h} f(s, y(s)) \, ds \approx h f(t_n, y_n).$$

Substituting this approximation gives Euler's update rule:

$$y_{n+1} = y_n + h f(t_n, y_n).$$

This is not merely inspired by integration. It *is* numerical integration. Each Euler step performs the following process:

1. Measure the current rate of change,
2. assume it remains roughly constant over a short interval,
3. accumulate the resulting small change,
4. repeat.

Thus, every Euler step is a discrete version of the net change theorem:

$$\text{new value} = \text{old value} + \text{accumulated change}.$$

This explains why:

- numerical ODE solving is often called *integration*,
- solution curves are called *integral curves*,
- ODE solvers are often called *integrators*.

3 Cauchy Formula for Repeated Integration

3.1 Basic understanding

1. What problem does the Cauchy formula for repeated integration solve?
2. What is meant by the " n -th repeated integral" of a function?
3. Write the definition of the n -fold repeated integral of a function f with the base point a :

$$f^{(-n)}(x) = ?$$

4. Under what conditions on f does the formula hold?
5. What is the main result of the Cauchy formula for repeated integration? State it explicitly.

3.2 Formula and structure

6. Rewrite the repeated integral as a single integral:

$$f^{(-n)}(x) = \frac{1}{(n-1)!} \int_a^x (x-t)^{n-1} f(t) dt.$$

7. What is the role of the factor $\frac{1}{(n-1)!}$?
8. Why does the kernel $(x-t)^{n-1}$ appear in the formula?
9. What happens in the special case $n = 1$? Verify that the formula reduces correctly.
10. Compute explicitly $f^{(-2)}(x)$ using both:
 - the definition (nested integrals),
 - the Cauchy formula,and verify that they agree.

3.3 Proof understanding

11. What proof technique is used to prove the formula?
12. What is the base case of the induction?
13. Show that for $n = 1$:

$$f^{(-1)}(x) = \int_a^x f(t) dt.$$

14. What assumption is made in the induction step?
15. What must be proven in the induction step?
16. Where is the Leibniz integral rule used in the proof?
17. Compute:

$$\frac{d}{dx} \left[\frac{1}{n!} \int_a^x (x-t)^n f(t) dt \right]$$

and explain why it gives the desired recursive relation.

18. How does the induction step transform an $(n+1)$ -fold integral into the desired single-integral form?
19. Why does the induction argument prove the formula for all $n \in \mathbb{N}$?

3.4 Conceptual understanding

20. Why can repeated integration be "compressed" into a single integral?
21. How is this formula related to convolution?
22. Interpret the kernel $(x - t)^{n-1}$ geometrically or intuitively.
23. How does this formula simplify practical computations?
24. In what situations is this formula especially useful?

3.5 Generalizations

25. How can the formula be extended to non-integer orders of integration?
26. What replaces the factorial $(n - 1)!$ in the generalized version?
27. State the fractional integral version:

$$(J^\alpha f)(x) = ?$$

28. What condition must α satisfy?
29. What is the name of this generalization?
30. How does the formula relate to fractional derivatives (when $\alpha < 0$)?

3.6 Connections and further ideas

31. How is this formula connected to fractional calculus?
32. What is a differintegral?
33. How can repeated integration be used to define differentiation of non-integer order?
34. How is this formula generalized to higher dimensions?
35. What is the Riesz potential?

3.7 Practice problems

36. Compute:

$$\int_0^x \int_0^{t_1} t_2 \, dt_2 \, dt_1$$

using the Cauchy formula.

37. Compute the n -fold integral of $f(t) = 1$.
38. Compute the n -fold integral of $f(t) = t^k$.
39. Show that:

$$(J^n f)(x) = (f * g_n)(x)$$

for an appropriate kernel g_n .

40. Verify the formula numerically for a simple function (e.g. $f(t) = e^t$).

4 Differentiation Under the Integral Sign

1. What is the statement of Leibniz's rule for differentiating under the integral sign with constant limits?
2. What is the general form of Leibniz's rule when the limits of integration depend on the parameter?
3. What conditions must be satisfied to justify differentiation under the integral sign (continuity, uniform convergence, dominated convergence, etc.)?
4. What role does continuity of the integrand and its partial derivative play in applying differentiation under the integral sign?
5. How does the Dominated Convergence Theorem justify differentiation under the integral sign?
6. When can uniform convergence be used to justify differentiation under the integral sign?
7. What is the difference between pointwise convergence and uniform convergence in this context?
8. How do you compute $\frac{d}{d\alpha} \int_a^b f(x, \alpha), dx$?
9. How do you compute $\frac{d}{d\alpha} \int_{a(\alpha)}^{b(\alpha)} f(x, \alpha), dx$?
10. What additional terms appear when the limits of integration depend on the parameter?
11. How do you apply differentiation under the integral sign to evaluate difficult integrals?
12. What is the idea behind Feynman's technique for evaluating integrals using parameters?
13. How can introducing a parameter simplify the evaluation of an integral?
14. Give an example where differentiating under the integral sign transforms an integral into a simpler one.
15. How do you recover the original integral after differentiating with respect to a parameter?
16. What initial or boundary conditions are needed after integrating back with respect to the parameter?
17. How do you handle constants of integration when using this method?
18. When is it valid to interchange differentiation and integration?
19. What are common pitfalls when applying differentiation under the integral sign?
20. How does this technique apply to improper integrals?
21. What extra care is needed when the interval of integration is infinite?
22. How do you justify differentiation under the integral sign for $\int_0^\infty f(x, \alpha), dx$?
23. Can differentiation under the integral sign be applied multiple times? Under what conditions?
24. What is the relationship between differentiation under the integral sign and parameter-dependent integrals?
25. How does this method connect to solving differential equations?
26. How can this technique be used in probability theory (e.g., moment generating functions)?

27. How does differentiation under the integral sign relate to Fourier transforms or Laplace transforms?
28. What are some classic integrals that are evaluated using this method?
29. How would you construct your own parameterized integral to apply this technique?
30. How do you verify that your final answer is correct after applying the method?

5 The Gamma Function

1. What is the definition of the Gamma function $\Gamma(x)$ for $x > 0$?
2. For which values of x is the integral definition of $\Gamma(x)$ valid?
3. Show that $\Gamma(1) = 1$.
4. Prove the functional equation $\Gamma(x+1) = x\Gamma(x)$.
5. How does the Gamma function extend the factorial function? State the relationship between $\Gamma(n)$ and $(n-1)!$ for $n \in \mathbb{N}$.
6. Compute $\Gamma(2)$, $\Gamma(3)$, and $\Gamma(4)$ using the functional equation.
7. What is the value of $\Gamma\left(\frac{1}{2}\right)$?
8. How can the integral $\int_0^\infty e^{-t^2} dt$ be used to compute $\Gamma\left(\frac{1}{2}\right)$?
9. What substitution transforms the Gaussian integral into the Gamma function integral?
10. What is the reflection formula for the Gamma function?
11. Use the reflection formula to compute $\Gamma\left(\frac{1}{2}\right)$.
12. What is the duplication formula (Legendre's formula) for the Gamma function?
13. What is the behavior of $\Gamma(x)$ near $x = 0$?
14. Does the Gamma function have any poles? If so, where are they located?
15. Is the Gamma function defined for negative non-integer values? What happens at negative integers?
16. What is the logarithmic derivative of the Gamma function called?
17. Define the digamma function $\psi(x)$.
18. What is the integral representation of $\Gamma(x)$ involving $e^{-t}t^{x-1}$?
19. What is an alternative integral representation of $\Gamma(x)$ using a limit (Euler's limit form)?
20. What is Stirling's approximation for $\Gamma(x)$ for large x ?
21. How can Stirling's formula be used to approximate $n!$?
22. What is the Beta function $B(x, y)$?
23. How is the Beta function related to the Gamma function?
24. Show that $B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$.
25. What is the integral definition of the Beta function?
26. How can you derive the Beta-Gamma relationship using a change of variables?
27. What is the asymptotic growth rate of $\Gamma(x)$ as $x \rightarrow \infty$?
28. Is the Gamma function convex or concave on $(0, \infty)$?
29. What is the logarithmic convexity (log-convexity) property of $\Gamma(x)$?
30. State Bohr-Mollerup theorem and explain its significance.
31. How does the Gamma function behave under scaling transformations?
32. What is the value of $\Gamma\left(\frac{3}{2}\right)$?

33. Compute $\Gamma\left(\frac{5}{2}\right)$ using recursion.
34. Express $\Gamma\left(n + \frac{1}{2}\right)$ in terms of factorials and $\sqrt{\pi}$.
35. What is the relation between $\Gamma(x)$ and improper integrals in probability theory?
36. How does the Gamma function appear in the definition of the Gamma distribution?
37. What is the probability density function of the Gamma distribution?
38. What role does $\Gamma(x)$ play in normalizing probability distributions?
39. How can $\Gamma(x)$ be analytically continued to complex numbers?
40. Is $\Gamma(x)$ an entire function? Why or why not?
41. Where are the zeros of the Gamma function located?
42. What is the Weierstrass product representation of $\Gamma(x)$?
43. How does $\Gamma(x)$ relate to complex analysis and contour integrals?
44. What is Euler's constant γ and how does it relate to $\Gamma(x)$?
45. What is the series expansion of $\log \Gamma(x)$ near $x = 1$?
46. How is the Gamma function used in evaluating definite integrals?
47. Give an example of an integral that evaluates to a Gamma function.
48. How does the substitution $t = -\ln u$ relate the Gamma function to integrals over $(0, 1)$?
49. What is the connection between $\Gamma(x)$ and factorial moments?
50. Why is the Gamma function important in physics and engineering?

6 Derivative of the Inverse Function

1. What is the definition of the inverse of a function f ? Under what condition does an inverse exist?
2. What does it mean for a function to be one-to-one (injective), and why is this property necessary for the existence of an inverse?
3. State the relationship between a function and its inverse using composition.
4. If $y = f(x)$, how do you express x in terms of f^{-1} ?
5. What is the geometric relationship between the graphs of f and f^{-1} ?
6. How can you determine whether a function has an inverse using the horizontal line test?
7. State the formula for the derivative of an inverse function:

$$(f^{-1})'(x) = ?$$

8. Derive the formula for $(f^{-1})'(x)$ starting from the identity $f(f^{-1}(x)) = x$.
9. If $f(a) = b$, express $(f^{-1})'(b)$ in terms of $f'(a)$.
10. Under what conditions on f does the derivative of the inverse function exist?
11. Why must $f'(a) \neq 0$ for the inverse function to be differentiable at $b = f(a)$?
12. Compute $(f^{-1})'(x)$ when $f(x) = x^3 + x$.
13. Let $f(x) = \sin x$ restricted to $[-\frac{\pi}{2}, \frac{\pi}{2}]$. What is $f^{-1}(x)$ and its derivative?
14. Find the derivative of $\arctan(x)$ using the inverse function derivative formula.
15. Show that the derivative of $\ln x$ can be derived from the inverse of e^x .
16. If $f'(x)$ is known, describe the steps to compute $(f^{-1})'(x)$ at a specific point.
17. What happens to $(f^{-1})'(x)$ if $f'(x)$ is very large? What is the geometric interpretation?
18. What happens to $(f^{-1})'(x)$ if $f'(x)$ is very small (but nonzero)?
19. Explain why the slopes of inverse functions at corresponding points are reciprocals.
20. If f is decreasing and invertible, what can you say about the sign of $(f^{-1})'(x)$?
21. Let $f(x) = x^5$. Compute $(f^{-1})'(x)$ explicitly.
22. If $f(x) = \tan x$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$, compute the derivative of its inverse.
23. Verify that $\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$ using the inverse function rule.
24. How does implicit differentiation relate to finding the derivative of an inverse function?
25. If f is differentiable and strictly monotonic on an interval, what can be said about the differentiability of f^{-1} ?
26. Give an example of a function that is invertible but whose inverse is not differentiable at some point.
27. What is the domain of f^{-1} in terms of the range of f ?
28. What is the range of f^{-1} in terms of the domain of f ?
29. How can you use the inverse function derivative formula to avoid explicitly finding the inverse?

30. Let $f(2) = 5$ and $f'(2) = 3$. Find $(f^{-1})'(5)$.

7 Newton's (Newton–Raphson) Method

1. What is the goal of a root-finding algorithm?
2. State the Newton method iteration formula for finding a root of a function $f(x)$.
3. How do you obtain the next iterate x_{n+1} from the tangent line?
4. Derive the Newton iteration by applying a first-order Taylor expansion of $f(x)$ around x_n .
5. What assumptions about $f(x)$ are required for Newton's method to work?
6. What is meant by an initial guess x_0 , and why is it important?
7. Define what it means for Newton's method to converge.
8. What is quadratic convergence?
9. Why does Newton's method converge faster near the root than far from it?
10. What happens if $f'(x_n) = 0$ during the iteration?
11. Why is Newton's method considered a local method?
12. Why can Newton's method fail to converge for some initial guesses?
13. Give an example of a function and initial guess where Newton's method diverges.
14. What is meant by the basin of attraction of a root?
15. How does the choice of initial guess affect which root Newton's method converges to?
16. What is a multiple root, and how does it affect convergence of Newton's method?
17. How can Newton's method be modified to handle multiple roots?
18. What stopping criteria can be used to terminate Newton's method?
19. Compare stopping criteria based on $|x_{n+1} - x_n|$ and $|f(x_n)|$.
20. How is Newton method formulated in terms of a vector-valued function $F(x)$?
21. How can Newton's method be extended to systems of nonlinear equations?
22. How do we define the Jacobian matrix $J_F(x)$ for a function $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$?
23. Write the Newton iteration formula for solving $F(x) = 0$ in $\mathbb{R}^n$.
24. What role does the Jacobian matrix play in multidimensional Newton's method?
25. What issues arise when the Jacobian is singular or nearly singular?
26. What is a damped (or modified) Newton method?
27. Why might damping improve convergence?
28. How can Newton's method be used to find extrema of a function?
29. How is Newton's method related to solving $f'(x) = 0$?
30. What is the difference between Newton's method for optimization and for root finding?
31. What is the order of convergence of Newton's method compared to secant and bisection methods?
32. What is the secant method, and how does it approximate Newton's method?
33. When might the secant method be preferred over Newton's method?

34. How can automatic differentiation help in implementing Newton's method?
35. What is line search in the context of Newton's method?
36. How can Newton's method behave near inflection points?
37. What happens if the function is not differentiable at the root?
38. Why can Newton's method oscillate between points instead of converging?
39. How can you detect divergence in practice?
40. What are hybrid methods that combine Newton's method with bracketing methods?
41. Why are hybrid methods often preferred in practice?
42. What is the role of scaling in Newton's method for systems?
43. What is meant by the error $e_n = x_n - r$, and how does it evolve in Newton's method?
44. Derive the error recurrence relation for Newton's method near a simple root.
45. What is the goal of Newton's method in the context of optimization?
46. How do we derive Newton's method for finding a local minimum of a scalar function $f(x)$?
47. What role does the gradient $\nabla f(x)$ play in Newton's method?
48. What is the Hessian matrix $\nabla^2 f(x)$ and how is it defined?
49. Write the Newton iteration formula for optimization in $\mathbb{R}^n$.
50. How does Newton's method extend to complex-valued functions?
51. What are basins of attraction in the complex plane?
52. How can we visualize the convergence behavior of Newton's method in the complex plane?
53. How do Newton fractals arise from iterating Newton's method in $\mathbb{C}$?
54. Why do the boundaries between basins of attraction exhibit fractal structure?
55. What are fixed points of the Newton iteration map?
56. Are all fixed points of the Newton map roots of $f(z)$?
57. What are attracting, repelling, and neutral fixed points in this context?

8 Euler's Method

8.1 Fundamental Concepts

1. What problem does Euler's method aim to solve? State the general form of an initial value problem.
2. What is meant by a numerical approximation to a solution of an ODE?
3. What role does the step size h play in Euler's method?

8.2 Geometric and Intuitive Understanding

4. What is the geometric interpretation of Euler's method in terms of tangent lines?
5. How would you visualize the steps of Euler's method on a slope field?

8.3 Derivation and Formulation

6. Starting from the differential equation $y'(x) = f(x, y)$, derive the Euler update formula.
7. Write the recursive formula for Euler's method, defining all variables involved.
8. Given an initial condition $y(x_0) = y_0$, how do you compute y_1 using Euler's method?
9. How do you compute y_{n+1} from y_n in Euler's method?

8.4 Error Analysis and Accuracy

10. What is the local truncation error in Euler's method?
11. What is the global truncation error in Euler's method?
12. What is the order of accuracy of Euler's method?
13. How does decreasing the step size h affect the accuracy of Euler's method?
14. What is the trade-off between accuracy and computational cost in Euler's method?
15. What types of errors can accumulate in Euler's method over many steps?
16. How can you estimate the error in Euler's method during computation?

8.5 Stability and Limitations

17. What is meant by stability in the context of numerical methods for ODEs?
18. Why can Euler's method become unstable for certain differential equations?
19. Give an example of a differential equation where Euler's method performs poorly.
20. What is a stiff differential equation, and why is Euler's method problematic for such equations?
21. What happens if the function $f(x, y)$ is not smooth? How does this affect Euler's method?
22. What are common pitfalls when using Euler's method in practice?

8.6 Variants and Improvements

23. What is the difference between explicit Euler and implicit Euler methods?
24. Write the update formula for the implicit Euler method.
25. What advantage does the implicit Euler method have over the explicit version?
26. How can you modify Euler's method to improve accuracy (e.g., midpoint or improved Euler methods)?
27. What is the idea behind the Heun method (improved Euler method)?

8.7 Theoretical Connections

28. How does Euler's method relate to the Taylor series expansion of the solution?
29. What term(s) of the Taylor series does Euler's method include or neglect?

8.8 Extensions and Generalizations

30. Can Euler's method be used for systems of differential equations? How?
31. How does Euler's method behave when applied backward in time (negative step size)?

8.9 Euler's Method as Integration

32. What is the standard form of an ordinary differential equation involving a function $y(t)$?
33. How can the differential equation $\frac{dy}{dt} = f(t, y)$ be rewritten as an equivalent integral equation?
34. What does the integral $\int_{t_n}^{t_{n+1}} f(s, y(s)) ds$ represent conceptually?
35. Why is the integral form of the solution exact, even though we may not know $y(s)$?
36. What is the main difficulty in computing the integral $\int_{t_n}^{t_{n+1}} f(s, y(s)) ds$ exactly?
37. What approximation is made in the Forward Euler method for the integrand $f(s, y(s))$?
38. What is the geometric interpretation of approximating $f(s, y(s))$ by $f(t_n, y(t_n))$?
39. How does this approximation simplify the integral over the interval $[t_n, t_{n+1}]$?
40. How do we define the step size h , and what role does it play?
41. How do we derive the Forward Euler update formula from the approximated integral?
42. Why can we interpret the Forward Euler method as a left-endpoint Riemann sum?
43. In what sense is Forward Euler performing "integration" of the differential equation?
44. Why does assuming a constant slope over the interval introduce error?

9 Roots of Unity

1. What is meant by an n th root of unity?
2. Write the equation whose solutions are the n th roots of unity.
3. How many distinct n th roots of unity are there? Why?
4. Express the n th roots of unity in exponential form using Euler's formula.
5. Write the general formula for the k th n th root of unity.
6. What is the principal (or primitive) n th root of unity?
7. How can all n th roots of unity be generated from a single primitive root?
8. What is the difference between a root of unity and a primitive root of unity?
9. State the condition under which a root of unity is primitive.
10. How many primitive n th roots of unity are there? Which function counts them?
11. What is Euler's totient function $\varphi(n)$ and how does it relate to roots of unity?
12. Represent the n th roots of unity geometrically in the complex plane.
13. What is the magnitude and argument of each n th root of unity?
14. How are the n th roots of unity distributed on the complex unit circle?
15. What is the angle between consecutive n th roots of unity?
16. Compute the sum of all n th roots of unity. What is the result?
17. Why does the sum of all n th roots of unity equal zero?
18. Compute the product of all n th roots of unity.
19. What symmetry properties do the n th roots of unity exhibit?
20. Show that the n th roots of unity form a group under multiplication.
21. What type of group is formed by the n th roots of unity?
22. What is the identity element in this group?
23. What is the inverse of a given n th root of unity?
24. Show that the set of n th roots of unity is cyclic.
25. What is a generator of the group of n th roots of unity?
26. For which values of k is ζ^k a primitive root, where ζ is a primitive n th root?
27. Solve the equation $z^n = 1$ explicitly for small values such as $n = 2, 3, 4$.
28. What are the cube roots of unity? Write them explicitly.
29. What are the fourth roots of unity? Write them explicitly.
30. How can roots of unity be used to factor the polynomial $x^n - 1$?
31. Write the factorization of $x^n - 1$ in terms of its complex roots.
32. What are cyclotomic polynomials and how are they related to roots of unity?
33. Define the n th cyclotomic polynomial $\Phi_n(x)$.
34. How does $x^n - 1$ factor into cyclotomic polynomials?

35. What is the relationship between primitive roots of unity and cyclotomic polynomials?
36. How can roots of unity be used to evaluate sums such as $\sum_{k=0}^{n-1} \zeta^k$?
37. Evaluate $\sum_{k=0}^{n-1} \zeta^{mk}$ for an integer m .
38. Under what condition does $\sum_{k=0}^{n-1} \zeta^{mk} = 0$?
39. What happens when m is a multiple of n in the above sum?
40. How are roots of unity used in discrete Fourier transforms (DFT)?
41. What is the connection between roots of unity and periodicity?
42. How can roots of unity simplify computations involving periodic sums?
43. Show that if ζ is an n th root of unity, then $\bar{\zeta} = \zeta^{-1}$.
44. What is the complex conjugate of a root of unity geometrically?
45. How do roots of unity relate to rotations in the complex plane?
46. What is the minimal polynomial of a primitive n th root of unity?
47. How does the degree of $\Phi_n(x)$ relate to $\varphi(n)$?
48. In what fields do primitive n th roots of unity lie?
49. What is the significance of roots of unity in number theory?
50. How are roots of unity used in solving polynomial equations?

10 Complex Numbers as Matrices / Representation Theory

<https://www.youtube.com/watch?v=hsveVFoIJPM>

1. What does it mean to represent an abstract algebraic object (like a group element or number) as a matrix?
2. What is the general idea of a **representation** in the context of linear algebra and group theory?
3. How can a complex number $a + bi$ be represented as a 2×2 real matrix?
4. Write explicitly the matrix corresponding to the complex number $a + bi$.
5. Show how matrix addition corresponds to addition of complex numbers under this representation.
6. Show how matrix multiplication corresponds to multiplication of complex numbers under this representation.
7. Why is this matrix representation of complex numbers structure-preserving?
8. What algebraic structure is preserved when mapping complex numbers to matrices?
9. What property must a mapping satisfy to be considered a homomorphism?
10. Verify that the mapping from complex numbers to matrices is a homomorphism.
11. What is the identity element in the complex numbers, and what matrix represents it?
12. What matrix corresponds to the imaginary unit i ?
13. Compute the square of the matrix representing i and interpret the result.
14. How does this matrix representation encode the fact that $i^2 = -1$?
15. What geometric transformation in $\mathbb{R}^2$ corresponds to multiplication by i ?
16. How does multiplication by a general complex number $a + bi$ act geometrically on the plane?
17. How is rotation represented in terms of matrices?
18. How does scaling appear in the matrix representation of a complex number?
19. Explain how complex multiplication combines rotation and scaling.
20. What is the determinant of the matrix corresponding to $a + bi$?
21. What is the geometric meaning of this determinant?
22. What is the trace of the matrix representation of $a + bi$?
23. How does the modulus $|a + bi|$ relate to the determinant of the matrix?
24. What condition on a and b makes the matrix invertible?
25. What is the matrix corresponding to the inverse of a complex number?
26. Show that the inverse matrix corresponds to the multiplicative inverse of the complex number.
27. How does this representation help visualize complex numbers as linear transformations?
28. What is the connection between this representation and rotations in the plane?

29. How can this idea be generalized to represent other algebraic structures using matrices?
30. What is a group representation in general?
31. Why are matrices particularly useful for representing abstract algebraic objects?
32. How does this example motivate the study of representation theory?
33. What advantages do matrix representations provide for computation?
34. How can eigenvalues of these matrices be interpreted in terms of complex numbers?
35. What happens when you diagonalize the matrix corresponding to a complex number?
36. How does this representation connect linear algebra with complex analysis?
37. In what sense are complex numbers “the same” as a subset of 2×2 real matrices?
38. What is the dimension of the vector space in which these matrices act?
39. How does this perspective change your understanding of multiplication by complex numbers?
40. Can every linear transformation of $\mathbb{R}^2$ be represented by a complex number? Why or why not?
41. What distinguishes matrices that correspond to complex numbers from arbitrary 2×2 matrices?
42. How would you test whether a given 2×2 matrix corresponds to a complex number?
43. What deeper insight does this example give about the relationship between algebra and geometry?

11 Hidden Constants in Integration by Parts

<https://www.youtube.com/watch?v=N5xtRr0jfWI>

<https://www.youtube.com/watch?v=V8uZYdd4Jnk>

1. Why does every indefinite integral include an arbitrary constant C ?
2. Derive the integration by parts formula from the product rule.
3. Why is the equation

$$\int u \, dv = uv - \int v \, du$$

technically incomplete if we ignore constants of integration? Where are the hidden constants in this equation?

4. Why can ignoring arbitrary constants sometimes lead to paradoxes or false identities when manipulating indefinite integrals?
5. Consider this example

$$I = \int \underbrace{\frac{f'(x)}{f(x)}}_{uv'} dx = \underbrace{\frac{f(x)}{f(x)}}_{uv} - \int \underbrace{-\frac{f'(x)}{f^2(x)}}_{u'} \underbrace{f(x)}_v dx = 1 + \int \frac{f'(x)}{f(x)} dx = 1 + I$$

$$\begin{aligned} u &= \frac{1}{f(x)} & v' &= f'(x) \\ u' &= -\frac{f'(x)}{f(x)^2} & v &= f(x) \end{aligned}$$

Where is the error in this paradoxical derivation?

6. Consider this example

$$I = \int f(x)g'(x) \, dx$$

$$\begin{aligned} u &= f(x) & v' &= g'(x) \\ u' &= f'(x) & v &= g(x) + A \end{aligned}$$

Show how all the constants of integration are absorbed into one when applying integration by parts.

7. Why can repeated applications of integration by parts accumulate multiple constants?
8. How can these multiple constants always be absorbed into a single constant in the final answer?
9. Why is it sometimes useful to deliberately choose a non-zero constant when computing an antiderivative inside integration by parts?
10. If the integration by part is applied naively to integral $\int x \arctan(x) \, dx$ it leads to an incomplete solution

$$\int \underbrace{x}_{v'} \underbrace{\arctan(x)}_u dx = \underbrace{\frac{x^2}{2}}_v \underbrace{\arctan(x)}_u - \int \underbrace{\frac{x^2}{2}}_v \underbrace{\frac{1}{x^2+1}}_{u'} dx.$$

How can choosing a convenient constant in the antiderivative simplify the naive solution?

11. If the integration by part is applied naively to integral $\int \ln(x+2) dx$ it leads to an incomplete solution

$$\int \underbrace{1}_{v'} \underbrace{\ln(x+2)}_u dx = \underbrace{x}_v \underbrace{\ln(x+2)}_u - \int \underbrace{x}_v \underbrace{\frac{1}{x+2}}_{u'} dx.$$

How can choosing a convenient constant in the antiderivative simplify the naive solution?

12. Why are definite integrals immune to the constant-of-integration ambiguity?

12 Geometric Meaning of the Third Derivative

<https://www.youtube.com/watch?v=SovllrJUQ64>

1. What is the geometric meaning of the first derivative $f'(x)$ in terms of the graph of $f(x)$?
2. How can you interpret the second derivative $f''(x)$ geometrically?
3. What does it mean for $f''(x) > 0$ in terms of the shape of the graph?
4. What does it mean for $f''(x) < 0$?
5. Define an inflection point in terms of concavity.
6. What condition on $f''(x)$ is necessary for an inflection point?
7. Why is $f''(x) = 0$ not sufficient to guarantee an inflection point?
8. What additional condition ensures that a point where $f''(x) = 0$ is actually an inflection point?
9. How can sign changes of $f''(x)$ be used to detect inflection points?
10. What is the geometric meaning of the third derivative $f'''(x)$?
11. How can $f'''(x)$ be interpreted as a rate of change of another geometric quantity?
12. If $f'''(x) > 0$, what does this say about how concavity is changing?
13. If $f'''(x) < 0$, what does this imply about the change in concavity?
14. How does $f'''(x)$ describe the “bending behavior” of a curve beyond concavity?
15. Explain how the third derivative relates to the steepness of the tangent slope.
16. How does the graph of $f'(x)$ help visualize $f''(x)$?
17. How does the graph of $f''(x)$ help visualize $f'''(x)$?
18. If $f''(x)$ has a local maximum or minimum, what can you say about $f'''(x)$ at that point?
19. What is the relationship between $f'''(x) = 0$ and extrema of $f''(x)$?
20. Can $f'''(x) = 0$ correspond to an inflection point of $f(x)$? Explain.
21. How would you distinguish between:
 - an inflection point of $f(x)$
 - an extremum of $f''(x)$
22. Describe how successive derivatives correspond to successive “rates of change” geometrically.
23. Give a physical interpretation of $f'(x)$, $f''(x)$, and $f'''(x)$ in motion (position, velocity, acceleration, ...).
24. What is the physical meaning of the third derivative in kinematics (often called jerk)?
25. How does the sign of the third derivative affect motion in a physical system?
26. Construct an example function where $f''(x) = 0$ but there is no inflection point.
27. Construct an example where $f'''(x) = 0$ but $f''(x)$ does not change sign.
28. Given a graph, how would you visually estimate where $f'''(x)$ is positive or negative?
29. How can you use higher derivatives to understand increasingly subtle features of a curve?

30. Why is the third derivative rarely discussed compared to the first and second in basic calculus?
31. In what types of problems does understanding the third derivative become important?

13 Second Derivative Test and Hessian

<https://www.youtube.com/watch?v=z0bSdV8DA4M>

13.1 Core Concept: What is the second derivative test really doing?

- What does the second derivative test attempt to determine about a function at a critical point?
- Why is checking only the first derivative insufficient for classifying critical points?
- In one variable, what does the sign of $f''(x)$ tell us geometrically?
- How can you interpret $f''(x)$ in terms of curvature?

13.2 From 1D to Multivariable Functions

- What is the analogue of the second derivative for multivariable functions?
- What is the Hessian matrix, and how is it constructed?
- Why can't we just look at "the second derivative" in multiple dimensions?
- What does the Hessian represent geometrically?

13.3 Quadratic Approximation

- What is the second-order Taylor expansion of a function near a critical point?
- Why does the linear term vanish at a critical point?
- What role does the quadratic term play in determining local behavior?
- How does the function locally resemble a quadratic form?

13.4 Quadratic Forms and Geometry

- What is a quadratic form?
- How can the expression $\mathbf{x}^T H \mathbf{x}$ describe local curvature?
- What does it mean for a quadratic form to be:
 - Positive definite?
 - Negative definite?
 - Indefinite?

13.5 Classification of Critical Points

- How does the definiteness of the Hessian classify a critical point?
- What condition corresponds to a local minimum?
- What condition corresponds to a local maximum?
- What condition corresponds to a saddle point?
- What happens if the Hessian is degenerate (determinant zero)?

13.6 Eigenvalues Interpretation

- Why are eigenvalues the key to understanding the Hessian?
- How do eigenvalues relate to curvature along different directions?
- What does it mean if all eigenvalues are positive?
- What does it mean if eigenvalues have mixed signs?
- How do eigenvectors relate to principal directions of curvature?

13.7 Geometric Intuition

- Why is a saddle point associated with “curving up in some directions and down in others”?
- How can you visualize curvature along different directions through a point?
- Why is the second derivative test really a statement about curvature in all directions?

13.8 Connection to Linear Algebra

- Why can any symmetric matrix be diagonalized?
- How does diagonalizing the Hessian simplify the quadratic form?
- What does the diagonal form reveal about the function locally?

13.9 Deeper Understanding

- Why is the second derivative test fundamentally about approximating the function by a quadratic surface?
- In what sense is the test coordinate-independent?
- How does this perspective generalize beyond $\mathbb{R}^2$ or $\mathbb{R}^3$?

13.10 Edge Cases and Limitations

- Why does the test fail when eigenvalues are zero?
- What additional analysis is needed in degenerate cases?
- Can higher-order terms change the classification?

14 Frullani's Integral

1. State Frullani's integral theorem.
2. What conditions on $f(x)$ are typically required for Frullani's theorem to hold?
3. Why must the limits $f(0)$ and $f(\infty)$ exist?
4. What is the role of the condition $a, b > 0$?
5. Why does the logarithm naturally appear in Frullani-type integrals?
6. Derive

$$\int_0^\infty \frac{e^{-ax} - e^{-bx}}{x} dx = \ln \frac{b}{a}.$$

7. Explain why

$$\int_0^\infty \frac{e^{-ax}}{x} dx$$

diverges.

8. Why does subtracting two individually divergent integrals sometimes produce a convergent result?
9. What happens if $a = b$?
10. What happens if one of the parameters tends to zero?
11. What happens if one of the parameters tends to infinity?
12. Show how Frullani's theorem can be derived using differentiation under the integral sign.
13. Define

$$F(a) = \int_0^\infty \frac{e^{-x} - e^{-ax}}{x} dx.$$

Compute $F'(a)$, use the initial condition $F(1) = 0$ to determine the constant of integration, and write down the closed form for $F(a)$.

14. When applying the Feynman trick to

$$I(a) = \int_0^\infty \frac{f(x) - f(ax)}{x} dx,$$

what initial condition pins down the constant of integration? Why does this condition hold?

15. Differentiate

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx$$

with respect to a , solve the resulting equation, and verify that the answer matches Frullani's theorem.

16. In the Feynman approach, differentiating with respect to a removes the $\frac{1}{x}$ factor from the integrand. Why does this make the integral tractable, and what condition on f justifies differentiating under the integral sign?
17. Why does the final answer depend only on $f(0)$ and $f(\infty)$ rather than the full shape of f ?
18. Give an example of a function for which Frullani's theorem fails, and identify which hypothesis is violated.

19. Derive Frullani's theorem from the identity

$$f(ax) - f(bx) = \int_b^a x f'(tx) dt.$$

20. Compute

$$\int_0^\infty \frac{\arctan(ax) - \arctan(bx)}{x} dx.$$

21. Compute

$$\int_0^\infty \frac{\ln(1+ax) - \ln(1+bx)}{x} dx.$$

Verify the answer by checking the $a = b$ limit.

22. Does Frullani's theorem apply to $\int_0^\infty \frac{\cos(ax) - \cos(bx)}{x} dx$? Identify which hypothesis is (or is not) satisfied, and compute the integral by another method.

23. What changes if the integration interval is $(0, T)$ rather than $(0, \infty)$?

24. Write $\frac{1}{x} = \int_0^\infty e^{-xs} ds$ and substitute into $\int_0^\infty \frac{f(ax) - f(bx)}{x} dx$. After a formal interchange of the order of integration, express the result in terms of the Laplace transform $\mathcal{L}\{f\}(s) = \int_0^\infty f(x)e^{-sx} dx$.

25. Show that

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = \int_0^\infty \left[\frac{1}{a} \mathcal{L}\{f\}\left(\frac{s}{a}\right) - \frac{1}{b} \mathcal{L}\{f\}\left(\frac{s}{b}\right) \right] ds.$$

Verify this for $f(x) = e^{-x}$ by computing the right-hand side directly and confirming the answer equals $\ln(b/a)$.

26. State the identity

$$\int_0^\infty \frac{g(x)}{x} dx = \int_0^\infty \mathcal{L}\{g\}(s) ds$$

and give sufficient conditions for its validity. Use it to rederive Frullani's theorem in one step by setting $g(x) = f(ax) - f(bx)$ and computing $\mathcal{L}\{g\}(s)$ explicitly.

27. In the Laplace-transform approach, why is the condition $g(0) = 0$ necessary for the identity in the previous question? Show that $g(0) = 0$ follows automatically when $g(x) = f(ax) - f(bx)$ and the hypotheses of Frullani's theorem are satisfied.
28. What justifies the interchange of the order of integration in the Laplace-transform proof? State the theorem you invoke (e.g. Tonelli's theorem) and verify its hypotheses for $f(x) = e^{-x}$.
29. How is Frullani's theorem connected to Laplace transforms in a broader sense? (*Hint*: consider the formula $\int_0^\infty f(x)/x dx = \int_0^\infty F(s) ds$, where $F = \mathcal{L}\{f\}$, and how it applies once the condition $g(0) = 0$ is ensured.)
30. Both the Feynman trick and the Laplace-kernel method deal with the $\frac{1}{x}$ factor. In the Feynman approach, $\frac{1}{x}$ is absorbed by differentiating with respect to a parameter; in the Laplace approach, $\frac{1}{x}$ is represented as $\int_0^\infty e^{-xs} ds$. Explain the structural analogy between these two approaches, and describe a setting in which one is preferable to the other.
31. What assumptions are required to differentiate under the integral sign? State a sufficient condition (e.g. dominated convergence) and verify it for $f(x) = e^{-x}$ in the Feynman proof of Frullani's theorem.

15 Dirichlet's Integral

1. State Dirichlet's integral:

$$\int_0^\infty \frac{\sin x}{x} dx.$$

What is its value?

2. Why is the integrand not actually singular at $x = 0$? Compute

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}.$$

3. Does

$$\int_0^\infty \left| \frac{\sin x}{x} \right| dx$$

converge? What does this tell you about the convergence of Dirichlet's integral?

4. Use integration by parts on

$$\int_1^R \frac{\sin x}{x} dx$$

to show that the limit as $R \rightarrow \infty$ exists.

5. For $a > 0$, compute

$$\int_0^\infty \frac{\sin(ax)}{x} dx$$

by the substitution $u = ax$.

6. What happens to the previous integral when $a = 0$?

7. What happens when $a < 0$? Use the oddness of $\sin$ to determine the sign.

8. Compare

$$\int_0^\infty \frac{\sin(ax)}{x} dx \quad \text{and} \quad \int_0^\infty \sin(ax) dx.$$

Why does the factor $\frac{1}{x}$ matter so much?

9. Define

$$F(a) = \int_0^\infty e^{-ax} \frac{\sin x}{x} dx, \quad a > 0.$$

Why is this integral easier to handle than

$$\int_0^\infty \frac{\sin x}{x} dx$$

directly?

10. Differentiate $F(a)$ with respect to a and show that

$$F'(a) = - \int_0^\infty e^{-ax} \sin x dx.$$

11. Compute

$$\int_0^\infty e^{-ax} \sin x dx$$

for $a > 0$.

12. Deduce that

$$F'(a) = -\frac{1}{1+a^2}.$$

Integrate this with respect to a .

13. Use the condition

$$\lim_{a \rightarrow \infty} F(a) = 0$$

to determine the constant of integration.

14. Show that

$$F(a) = \frac{\pi}{2} - \arctan a = \arctan \frac{1}{a}.$$

15. Let $a \rightarrow 0^+$ and recover Dirichlet's integral.

16. Why is introducing the factor e^{-ax} a useful trick before letting $a \rightarrow 0^+$?

17. Verify the identity

$$\frac{\sin x}{x} = \int_0^1 \cos(tx) dt.$$

18. Substitute this identity into

$$\int_0^\infty \frac{\sin x}{x} dx.$$

What formal double integral do you obtain?

19. Why is it dangerous to interchange the order of integration in that double integral without extra justification?

20. Explain why

$$F(a) = \int_0^\infty e^{-ax} \frac{\sin x}{x} dx$$

can be viewed as the Laplace transform of $\frac{\sin x}{x}$ evaluated at a .

21. If

$$f(x) = \frac{\sin x}{x},$$

write $F(a) = \mathcal{L}\{f\}(a)$. What does the formula

$$F(a) = \frac{\pi}{2} - \arctan a$$

tell you about $\mathcal{L}\left\{\frac{\sin x}{x}\right\}(a)$?

22. Starting from

$$\frac{\sin x}{x} = \int_0^1 \cos(tx) dt,$$

use linearity of the Laplace transform to derive

$$\mathcal{L}\left\{\frac{\sin x}{x}\right\}(a) = \int_0^1 \mathcal{L}\{\cos(tx)\}(a) dt.$$

Why is this a safer way to use the identity than working directly with

$$\int_0^\infty \frac{\sin x}{x} dx?$$

23. Compute

$$\mathcal{L}\{\cos(tx)\}(a)$$

for $a > 0$, and use it to show that

$$\mathcal{L}\left\{\frac{\sin x}{x}\right\}(a) = \int_0^1 \frac{a}{a^2 + t^2} dt.$$

In other words, show that

$$F(a) = \int_0^1 \int_0^\infty e^{-ax} \cos(tx) dx dt.$$

24. Evaluate

$$\int_0^1 \frac{a}{a^2 + t^2} dt$$

and recover the formula

$$\mathcal{L}\left\{\frac{\sin x}{x}\right\}(a) = \frac{\pi}{2} - \arctan a.$$

25. Use the Laplace-transform identity

$$\mathcal{L}\{f'\}(a) = a\mathcal{L}\{f\}(a) - f(0)$$

with

$$f(x) = \frac{\sin x}{x}$$

to explain informally why the transform of $\frac{\sin x}{x}$ should be related to an arctangent.

26. Why does the limit

$$\lim_{a \rightarrow 0^+} \mathcal{L}\left\{\frac{\sin x}{x}\right\}(a)$$

recover Dirichlet's integral?

27. Compare these two viewpoints: differentiating

$$F(a) = \int_0^\infty e^{-ax} \frac{\sin x}{x} dx$$

with respect to a , and recognizing $F(a)$ as a Laplace transform from the start. Why are these really two versions of the same idea?

28. Which route feels more natural if you already know Laplace transforms well: differentiating $F(a)$ first, or rewriting $F(a)$ as a Laplace transform of $\frac{\sin x}{x}$? Explain your answer.

29. Why is the value $\pi/2$ plausible from the formula

$$F(a) = \frac{\pi}{2} - \arctan a$$

when $a \rightarrow 0^+$?

30. Explain in words why oscillation of $\sin x$ is essential for convergence on $(1, \infty)$.

31. Sketch the graph of $\frac{\sin x}{x}$ and explain how the alternating positive and negative lobes fit the value of the integral.

32. Compute

$$\int_0^\infty \frac{\sin(bx)}{x} dx$$

for $b > 0$, and verify your formula by checking the special case $b = 1$.

16 Cauchy Principal Value

1. What problem is the Cauchy principal value trying to solve?
2. Why does

$$\int_{-1}^1 \frac{1}{x} dx$$

fail to exist as an ordinary improper integral?

3. What is the difference between an improper integral and a principal value?
4. Why is a principal value not simply another way to evaluate an improper integral?
5. What does the word “symmetric” mean in the definition of principal value?
6. Why is symmetry essential in the definition?
7. Can a principal value exist when the corresponding improper integral does not?
8. Can an improper integral exist when the principal value also exists?
9. What information is lost when we replace an improper integral by a principal value?
10. What geometric intuition can be attached to principal values?

16.1 Computational Questions

11. Compute

$$\text{PV} \int_{-1}^1 \frac{1}{x} dx.$$

12. Why does the answer not contradict the fact that the ordinary improper integral diverges?
13. Compute

$$\text{PV} \int_{-2}^3 \frac{1}{x} dx.$$

14. Why is the answer no longer zero?
15. Derive a general formula for

$$\text{PV} \int_a^b \frac{1}{x} dx,$$

assuming $a < 0 < b$.

16. Compute

$$\text{PV} \int_{-1}^1 \frac{x}{x^2 + 1} dx.$$

17. Compute

$$\text{PV} \int_{-1}^1 \frac{1+x}{x} dx.$$

18. Why is it useful to split the integrand into a singular part and a regular part?
19. Compute

$$\text{PV} \int_{-1}^1 \frac{\sin x}{x} dx.$$

20. Is a principal value always finite when the singularity is of the form $1/x$?

16.2 Questions About Singularities

21. Why does $1/x$ behave differently from $1/x^2$?
22. Does

$$\text{PV} \int_{-1}^1 \frac{1}{x^2} dx$$

exist?

23. If not, why not?
24. What property of $1/x$ allows cancellation?
25. What property of $1/x^2$ prevents cancellation?
26. For which powers p does

$$\text{PV} \int_{-1}^1 \frac{1}{x^p} dx$$

exist?

27. What role does oddness or evenness of a function play?
28. When does symmetry force a principal value to be zero?
29. Can an odd function have a divergent principal value?
30. Can an even function have a finite principal value?

16.3 Questions About Infinity

31. What is

$$\text{PV} \int_{-\infty}^{\infty} \frac{1}{x} dx?$$

32. How does this differ from

$$\int_{-\infty}^{\infty} \frac{1}{x} dx?$$

33. Why must limits at $+\infty$ and $-\infty$ be taken symmetrically?
34. What happens if the two limits are taken independently?
35. Give an example where symmetric truncation gives a finite value but ordinary convergence fails.
36. Why is

$$\text{PV} \int_{-\infty}^{\infty} \sin x dx$$

an interesting example?

37. How does principal value at infinity differ from principal value around an interior singularity?

16.4 Theory Questions

38. State the formal definition of the Cauchy principal value for an interior singularity.
39. State the formal definition for an integral over the whole real line.
40. Why is the principal value defined as a limit?

41. Under what conditions does the principal value agree with the ordinary improper integral?
42. Is the principal value linear?
43. When is it legitimate to split a principal value integral into two pieces?
44. Can principal values always be manipulated algebraically as ordinary integrals?
45. What pitfalls arise when changing variables in principal value integrals?
46. Why must one be careful about symmetry when performing substitutions?
47. Is principal value integration compatible with all standard integration theorems?
48. Which familiar theorems from elementary calculus continue to work unchanged?
49. Which require extra hypotheses?
50. Why do analysts treat principal values as generalized notions of integration?

17 Cauchy–Schwarz Inequality

17.1 Basic Definitions and Geometric Meaning

- State the Cauchy–Schwarz inequality for vectors in $\mathbb{R}^n$.
- State the inequality for arbitrary inner product spaces.
- What conditions must a function $\langle \cdot, \cdot \rangle$ satisfy to be an inner product?
- Explain geometrically what the inequality says about two vectors.
- Why does the inequality imply that

$$|\cos \theta| \leq 1$$

for the angle between vectors?

- Derive the formula

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}$$

from the inequality.

- Compute both sides of the inequality for

$$u = (1, 2), \quad v = (3, -4).$$

- Verify the inequality for

$$u = (1, 1, 1), \quad v = (2, 0, -1).$$

- What happens if one of the vectors is the zero vector?
- Why is the inequality trivial in that case?

17.2 Equality Conditions

- Under what conditions does equality hold in the Cauchy–Schwarz inequality?
- Prove that equality holds if $u = \lambda v$ for some scalar λ .
- Prove the converse: if equality holds, then the vectors are linearly dependent.
- Give examples of:
 1. vectors for which equality holds,
 2. vectors for which the inequality is strict.
- Explain geometrically why parallel vectors produce equality.
- Can orthogonal vectors produce equality? Why or why not?

17.3 Proof Techniques

- Prove the inequality using the nonnegativity of

$$\|u - tv\|^2.$$

- In that proof, why must the discriminant of the resulting quadratic polynomial be nonpositive?

- Expand

$$\|u - tv\|^2$$

explicitly and identify the quadratic coefficients.

- Give a proof using orthogonal projection.
- Give a proof using minimization.
- Why does positivity of norms play a central role in every proof?
- Which assumptions about the inner product are actually used in the standard proof?

17.4 Complex Inner Product Spaces

- State the Cauchy–Schwarz inequality for complex vector spaces.
- Why must absolute values appear around the inner product?
- What changes in the proof for complex spaces?
- Verify the inequality for

$$u = (1 + i, 2), \quad v = (i, 1 - i).$$

- Why is

$$\langle u, u \rangle$$

always real and nonnegative even in complex spaces?

- Explain why

$$\langle u, v \rangle \neq \langle v, u \rangle$$

in general for complex inner products.

17.5 Connections to Other Inequalities

- Derive the triangle inequality from Cauchy–Schwarz.
- Show how Cauchy–Schwarz implies

$$(a_1 + \cdots + a_n)^2 \leq n(a_1^2 + \cdots + a_n^2).$$

- Derive the inequality

$$\left(\sum_{k=1}^n a_k b_k \right)^2 \leq \left(\sum_{k=1}^n a_k^2 \right) \left(\sum_{k=1}^n b_k^2 \right).$$

17.6 Function Spaces and Integrals

- State the integral version of the Cauchy–Schwarz inequality.
- Verify the inequality for

$$f(x) = x, \quad g(x) = 1 - x$$

on $[0, 1]$.

- Show that

$$\left(\int_a^b f(x)g(x) dx \right)^2 \leq \left(\int_a^b f(x)^2 dx \right) \left(\int_a^b g(x)^2 dx \right).$$

- What inner product naturally appears in function spaces?
- What does equality mean in the integral version?
- Why is the integral version important in Fourier analysis?

17.7 Problem-Solving and Exploration

- Find vectors $u, v \in \mathbb{R}^2$ such that

$$\langle u, v \rangle = 0.$$

- Find nonzero vectors satisfying equality in Cauchy–Schwarz.
- Construct an example where the inequality is very far from equality.
- Use Cauchy–Schwarz to prove

$$(a + b + c)^2 \leq 3(a^2 + b^2 + c^2).$$

- Use the inequality to bound

$$|x + 2y + 3z|.$$

- Find the maximum possible value of

$$x + 2y$$

subject to

$$x^2 + y^2 = 1.$$

- Show that the angle definition from inner products is well-defined.
- Explain why projections minimize distance.
- Derive the least-squares approximation formula using orthogonality and Cauchy–Schwarz.
- Investigate whether analogous inequalities exist for norms that are not induced by inner products.